Inheritance Data: Predict, Calculate, Evaluate

LAUNCH · Science · 40 minutes

I can calculate inheritance probabilities, ratios and percentages and evaluate why observed results can differ from predictions.

Prepare before pupils arrive

Setup: Prepare the simulated 40-seedling dataset, repeated batches, blank grids and calculators. Check all arithmetic before teaching.

Safety: Keep the context botanical and fictional. Do not infer real genotypes or turn accuracy into public ranking.

Equipment: Blank Punnett grids, simulated repeated-batch dataset, calculator, formula strip, graph paper or comparison table and pencil.

Safety: Use fictional plant data only. No personal genotype, family history or medical-risk calculation is requested.

Fallback: The pupil selects the operation and dictates reasoning while an adult enters numbers, plots or scribes exact words.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Calculate and analyse outcomes using probabilities, ratios and percentages from monohybrid crosses.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	State the phenotype ratio for Pp × Pp. Convert 1/4 to a percentage.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	probability — chance of an outcome ratio — relative amounts sample size — number observed From Pp × Pp, predict 3/4 purple and 1/4 white: 75% and 25%. For 40 seedlings, calculate expected counts: 0.75 × 40 = 30 purple; 0.25 × 40 = 10 white.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Use repeated batches: individual groups can vary more than the combined total. A larger sample often gives an observed proportion closer to the expected probability, although exact agreement is not guaranteed.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Classify predictions, observations and valid conclusions, then repair the guarantee claim. PREDICT COUNT PERCENT COMPARE EVALUATE Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Complete a full inheritance data response: Punnett prediction, ratios, probabilities, observed percentages and evaluation. Record: Visible working from cross to percentages, a correct expected-observed comparison and an evaluation naming chance, sample size and one model limit.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Compare 77.5% observed with 75% predicted in one cautious sentence. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Observed 77.5% purple is close to predicted 75%; chance can explain a small difference.

Yes. The observed result is close to the expected proportion, and random sampling variation is normal.

Misconception repair

Repaired. Observed proportions can differ from expected probabilities, especially in finite samples; preserve the real data.

Guided checkpoint

Yes. The observed result is close to the expected proportion, and random sampling variation is normal. Compare percentages in the same unit and remember that probability is not a guarantee.

75% purple

EXPECTED FROM MODEL. Expected value placed. Ask whether this came from the square or the seedling counts.

31 purple of 40

OBSERVED IN DATA. Observed value placed. Ask whether this was actually counted in the dataset.

Close, not exact

CAUTIOUS CONCLUSION. Valid conclusion placed. Look for a sentence that compares in the same unit without claiming a guarantee.

Model limitation

The dataset is simulated, the inheritance model assumes complete dominance and one gene, and closeness alone cannot prove the biological assumptions.

Independent evidence

Visible working from cross to percentages, a correct expected-observed comparison and an evaluation naming chance, sample size and one model limit.

Supported: Complete a partly filled square, convert 3/4 and 1/4 to percentages, then choose the valid data conclusion. Formula and phenotype key stay visible; totals and first calculation are pre-positioned.

Standard: Build the square, calculate expected counts and observed percentages, then explain the small difference. Use Predict → Calculate → Compare → Explain.

Stretch: Analyse repeated batches and the combined sample, then evaluate sample size and the monohybrid model's limits. No chi-squared test or Higher-tier genetics; depth comes from careful evaluation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W13L3_Inheritance_Probability_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.